

외계행성 대기

Exoplanetary Atmospheres

Lecture 3: Radiative Transfer — Optical Depth, Opacity, and the Transfer Equation 2026 September 21 (Monday)

Based on: Chapters 2 (§2.5–2.9) and 5 of Seager, *Exoplanet Atmospheres* (2010), and Chapter 2 of Heng, *Exoplanetary Atmospheres: Theoretical Concepts and Foundations* (2017)

선훈광일 (Kwangil Seon)
KASI / UST

Outline – Lecture 3

Observed Flux at Earth

Luminosity, Incident Flux, and Blackbodies

Optical Depth: The Most Fundamental Quantity

Opacity: Absorption, Scattering, and Emission

Local Thermodynamic Equilibrium and the Source Function

The Equation of Radiative Transfer

Summary

Observed flux: what we measure at Earth

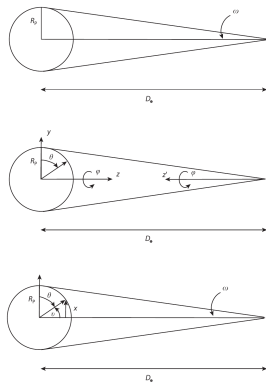


Figure 1: Derivation of the planet flux observed at Earth. Seager (2010), Fig. 2.5.

What radiative quantity can we measure at Earth? Because the planet is spatially unresolved, it appears as a point source: all radiation from its visible hemisphere is combined into a single measured flux, $F_{\oplus}(\nu, t)$.

- ▶ Recall that the surface intensity depends on the location on the planet, described by θ and ϕ . We must integrate the surface intensity from each location into a quantity we can actually measure.
- ▶ At Earth, a detector of a given area measures the energy collected from a planet that subtends a solid angle Ω , over specified frequency and time intervals.
- ▶ **The key point:** to derive $F_{\oplus}$ for a distant planet we must note that the integration over solid angle is performed *at the detector*, not at the planet surface.
- ▶ We return to the definition of flux and use the colatitude coordinate system, which is convenient here: the flux leaves the planet along the z -axis, and the z -axis points toward the detector at Earth. The angle ϕ is defined about the z -axis.

Observed flux: the solid angle subtended by the planet

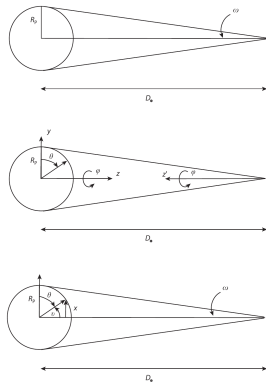


Figure 2: Top: the angle $R_p/D_\oplus$.
Middle: the shared azimuth ϕ .
Bottom: ω versus ϑ . Seager
(2010), Fig. 2.5.

- From the detector at Earth, the planet subtends the solid angle

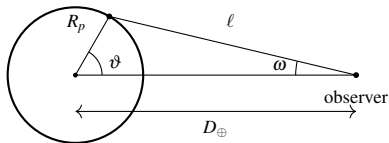
$$\Omega = \int_0^{2\pi} \int_0^{\omega_{\max}} \sin \omega \, d\omega \, d\phi \simeq \int_0^{2\pi} \int_0^{\omega_{\max}} \omega \, d\omega \, d\phi = \pi \left(\frac{R_p}{D_\oplus} \right)^2,$$

$$\sin \omega \simeq \omega, \quad \sin \omega_{\max} \simeq \omega_{\max} = R_p/D_\oplus,$$

where ω is the angle measured from the planet's center as seen from Earth (top panel), running from 0 at the disk center to $R_p/D_\oplus$ at the limb.

- A critical point: the azimuthal coordinate ϕ on the planet's apparent disk is *identical* to the coordinate ϕ in the planet's colatitude system (middle panel). The azimuth about the line of sight is the same whether measured at the planet or at the detector.
- Each point of the planet's disk, at angular position (ω, ϕ) as seen from Earth, corresponds to one surface element at colatitude ϑ on the planet (bottom panel); the ray from that element to Earth leaves the surface at angle ϑ from the local normal.

Observed flux: relating ω to the colatitude ϑ



A surface point at colatitude ϑ lies a distance ℓ from the observer and is seen at the angle ω from the line of sight. The cosine rule for the triangle gives

$$\ell = \sqrt{R_p^2 + D_\oplus^2 - 2R_p D_\oplus \cos \vartheta},$$

and the sine rule, applied to the same triangle,

$$\ell \sin \omega = R_p \sin \vartheta \quad \implies \quad \sin \omega = \frac{R_p}{\ell} \sin \vartheta.$$

Eliminating ℓ and dividing through by $D_\oplus$,

$$\sin \omega = \frac{(R_p/D_\oplus) \sin \vartheta}{\sqrt{1 + (R_p/D_\oplus)^2 - 2(R_p/D_\oplus) \cos \vartheta}}$$

which is exact. For a distant planet, $R_p/D_\oplus \ll 1$, the square root tends to 1 and

$$\sin \omega \simeq \frac{R_p}{D_\oplus} \sin \vartheta, \quad \text{so} \quad \omega \simeq \frac{R_p}{D_\oplus} \sin \vartheta$$

since ω is then itself small.

Observed flux: the integral at the detector

In the coordinate system of the detector, the flux is

$$F_{\oplus}(\mathbf{v}, t) = \int_0^{2\pi} \int_0^{R_p/D_{\oplus}} I_S(\vartheta, \phi, \mathbf{v}, t) \cos \omega \sin \omega \, d\omega \, d\phi,$$

where the intensity is still described in the planet's coordinates rather than in those of the detector. To convert $\cos \omega \sin \omega \, d\omega$ into the planet's coordinates, differentiate the small-angle relation of the previous slide:

$$\cos \omega \, d\omega \simeq \frac{R_p}{D_{\oplus}} \cos \vartheta \, d\vartheta,$$

and multiply by $\sin \omega \simeq (R_p/D_{\oplus}) \sin \vartheta$:

$$\cos \omega \sin \omega \, d\omega \simeq \left(\frac{R_p}{D_{\oplus}} \right)^2 \cos \vartheta \sin \vartheta \, d\vartheta$$

- The upper limit $\omega = R_p/D_{\oplus}$ corresponds to $\vartheta = \pi/2$, the limb of the visible hemisphere.
- In the same limit $D_{\oplus}$ may be taken as the distance from the surface of the star to the observer at Earth.

Substituting,

$$F_{\oplus}(\mathbf{v}, t) = \left(\frac{R_p}{D_{\oplus}} \right)^2 \int_0^{2\pi} \int_0^{\pi/2} I_S(\vartheta, \phi, \mathbf{v}, t) \cos \vartheta \sin \vartheta \, d\vartheta \, d\phi,$$

in which the angular integral is exactly the surface-flux integral.

Observed flux: the result

$$F_{\oplus}(\nu, t) = \left(\frac{R_p}{D_{\oplus}} \right)^2 F_S(\nu, t)$$

- ▶ $F_{\oplus}$ has the same dimensions as F_S and is smaller by the dilution factor $(R_p/D_{\oplus})^2$: the inverse-square law, now derived rather than assumed. For a Jupiter at 10 pc, $(R_p/D_{\oplus})^2 = 5 \times 10^{-17}$.
- ▶ The result is exact only for a planet with uniform surface intensity and $R_p \ll D_{\oplus}$. For a nonuniform planet, $F_{\oplus}$ is a hemisphere-weighted average with the $\cos \vartheta$ weighting: regions near the limb contribute less than regions near the center of the disk. This weighting is what phase curves and eclipse maps exploit to recover the brightness distribution.
- ▶ The surface flux and the observed flux are thus the “same” quantity up to a geometric factor, provided the planet is far away and its intensity is uniform. This is the justification for talking about the planet’s “flux” without specifying where it is evaluated, which we will often do.
- ▶ The same derivation with the roles reversed gives the stellar flux incident on the planet (next section).

Luminosity and outgoing energy

Luminosity $L(t)$ is the rate at which a planet radiates energy in *all* directions: the surface flux summed over a closed surface around the planet. For uniform flux $F_S(t)$ radiated equally everywhere, with $dA = R_p^2 \sin \theta d\theta d\phi$,

$$L(t) = \oint_A F_S(t) dA = R_p^2 \int_0^{2\pi} \int_0^\pi F_S(t) \sin \theta d\theta d\phi \implies \boxed{L(t) = 4\pi R_p^2 F_S(t)} \quad [\text{J s}^{-1}].$$

- For giant planets, the interior luminosity is also useful for determining the flux incident on the atmosphere's *lower* boundary. This flux is uniform across the planet and originates from gravitational energy released as the planet contracts.
- **Caution:** when observing, only *one hemisphere* is visible. For a planet that does not radiate uniformly (a hot dayside and a cold nightside) we must not use L ; instead use the energy per unit time leaving one hemisphere,

$$E_S(t) = R_p^2 \int_0^\pi \int_0^{\pi/2} F_S(\theta, \phi, t) \sin \theta d\theta d\phi \xrightarrow{\text{uniform}} E_S(t) = 2\pi R_p^2 F_S(t).$$

Energy per unit time is technically *power*; the planetary literature says “energy per unit time” or “luminosity”.

Incident flux: the star as seen from the planet (at substellar point)

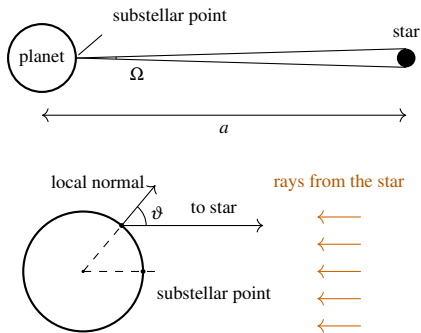


Figure 3: Top: incident flux at the substellar point, where the star subtends the solid angle Ω and a is the semimajor axis (after Seager 2010, Fig. 2.6). Bottom: at any other surface element the direction to the star makes the angle ϑ with the local surface normal.

Stellar radiation heats the planet, governs its global energy balance, and drives atmospheric circulation. We call it **incident radiation, incident flux, or irradiation**.

We can reuse the result for the planetary flux observed at Earth by reversing the roles: replace $R_p \rightarrow R_*$, $D_\oplus \rightarrow a$, and $I_S \rightarrow I_{S,*}$. The flux from the star at the substellar point is

$$F_{\text{inc}}(\nu, t) = \left(\frac{R_*}{a}\right)^2 \int_0^{2\pi} \int_0^{\pi/2} I_{S,*} \cos \vartheta_* \sin \vartheta_* d\vartheta_* d\phi_*.$$

For a star of uniform surface intensity (no limb darkening),

$$F_{\text{inc}}(\nu, t) = \left(\frac{R_*}{a}\right)^2 \pi I_{S,*}(\nu, t) = \left(\frac{R_*}{a}\right)^2 F_{S,*}(\nu, t)$$

This is the flux at the **substellar point**, where the star is at the zenith.

Incident energy: integrating over the dayside

The total incident energy per unit time is *not* $2\pi R_p^2 F_{\text{inc}}$: only the substellar point receives the full flux. A point at an angle ϑ from the substellar point receives $F_{\text{inc}} \cos \vartheta$ because the surface element is tilted (this is why Earth's poles are colder than the equator). Integrating over the illuminated hemisphere,

$$E_{\text{inc}}(\nu, t) = \left(\frac{R_*}{a}\right)^2 F_{\text{S},*}(\nu, t) \int_0^{2\pi} \int_0^{\pi/2} R_p^2 \cos \vartheta \sin \vartheta \, d\vartheta \, d\phi \quad \Rightarrow \quad E_{\text{inc}}(\nu, t) = \pi R_p^2 \left(\frac{R_*}{a}\right)^2 F_{\text{S},*}(\nu, t)$$

and, integrated over all wavelengths, $E_{\text{inc}}(t) = \pi R_p^2 (R_*/a)^2 F_{\text{S},*}(t)$.

- ▶ The $\cos \vartheta$ integral turns the hemisphere ($2\pi R_p^2$) into the **cross-sectional disk** πR_p^2 : the planet intercepts starlight with its projected area. This is the “disk” in Fig. 2.6 in the textbook.
- ▶ Sanity check with $L_* = 4\pi R_*^2 F_{\text{S},*}$: the stellar flux at distance a is $L_*/4\pi a^2 = (R_*/a)^2 F_{\text{S},*}$, and the planet intercepts πR_p^2 of it. For Earth, $L_{\odot}/4\pi a^2 = 1361 \text{ W m}^{-2}$ (the solar constant).

The star as a point source: plane-parallel incident radiation

For radiative-transfer calculations at a given surface element, the star is far enough away that only rays from *one* direction are incident. We then describe the incident stellar intensity with delta functions,

$$I_*(\vartheta, \phi, \nu, t) = I_0 \delta(\vartheta - \vartheta_0) \delta(\phi - \phi_0),$$

where (ϑ_0, ϕ_0) is the direction of the star from the local surface normal ($\vartheta = \vartheta_0$ in Figure 11); this direction differs from one surface element to another. The incident flux at the element is then

$$F_{\text{inc}}(\vartheta, \phi, \nu, t) = \left(\frac{R_*}{a}\right)^2 \int_0^{2\pi} \int_0^{\pi/2} I_0 \delta(\vartheta - \vartheta_0) \delta(\phi - \phi_0) \cos \vartheta \sin \vartheta \, d\vartheta \, d\phi = \left(\frac{R_*}{a}\right)^2 \cos \vartheta_0 \sin \vartheta_0 I_0.$$

- ▶ The $\cos \vartheta_0$ is the projection factor; $\sin \vartheta_0$ comes from the solid-angle Jacobian and the normalization of the delta functions in (ϑ, ϕ) .
- ▶ This “plane-parallel beam” form is the upper boundary condition of the plane-parallel radiative transfer equation (Chapter 5): $I(0, \mu, \nu) = I_* \delta(\mu - \mu_0)$ for incoming rays.
- ▶ *When is the point-source approximation valid?* (exercise 2.7) When the star’s angular radius R_*/a is small compared to the angular scales over which the atmosphere’s response varies. For a hot Jupiter at $a = 0.05$ AU, $R_*/a \simeq 0.09$ rad $\simeq 5^\circ$, so **near the terminator (the day–night boundary, where the star sits on the horizon and $\vartheta_0 \simeq 90^\circ$) the finite size of the star (and its limb darkening) already matters.**

Blackbody intensity and blackbody flux

A **blackbody** is a perfect radiator: it absorbs all incident radiation and re-emits it with a spectrum that depends only on its temperature T . Blackbody radiation is *isotropic* (no $\hat{n}$ -dependence). Its intensity is the **Planck function**

$$B(T, \nu) = \frac{2h\nu^3}{c^2} \frac{1}{e^{h\nu/k_B T} - 1} \quad [\text{J m}^{-2} \text{ sr}^{-1} \text{ s}^{-1} \text{ Hz}^{-1}],$$

the same dimensions as I . Since T varies with location (and possibly time) we may also write $B(\mathbf{x}, \nu)$.

- **Per wavelength.** Blackbody radiation per wavelength bin $d\lambda$ requires the conversion $d\nu = -(c/\lambda^2)d\lambda$ (the sign is absorbed into the direction of $d\lambda$):

$$B(T, \lambda) = \frac{2hc^2}{\lambda^5} \frac{1}{e^{hc/\lambda k_B T} - 1}, \quad B_\lambda d\lambda = B_\nu d\nu.$$

B_λ and B_ν peak at *different* locations (next slide), a common source of confusion (Heng, problem 2.8.4).

- **Blackbody flux.** Because B is isotropic, the flux outward through one hemisphere is

$$F_S(T, \nu) = \pi B(T, \nu)$$

We always use B for blackbody intensity and πB for blackbody flux.

Blackbody fluxes across the relevant temperatures

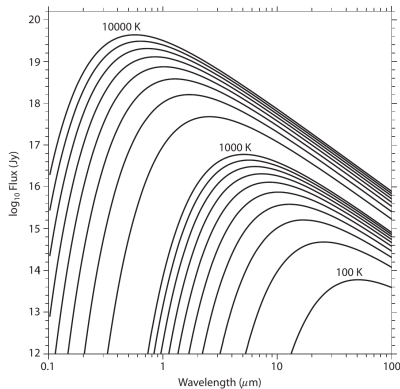


Figure 4: Blackbody surface fluxes πB (in $\text{Jy} = 10^{-26} \text{ W m}^{-2} \text{ Hz}^{-1}$), spaced by 100 K from 100 to 1000 K and by 1000 K from 1000 to 10,000 K. Seager (2010), Fig. 2.7.

- ▶ Planet-hosting stars typically have $T_{\text{eff}} \simeq 3000\text{--}6000 \text{ K}$ (the Sun may be represented by a 5750 K blackbody). Known planets have $T_{\text{eff}} \simeq 60$ to more than 2000 K (Earth is approximately a 300 K blackbody).
- ▶ Both the *magnitude* and the *frequency of the peak* of the blackbody flux increase with temperature; the curves never cross. A hotter body is brighter at *every* wavelength.
- ▶ Stellar and planetary atmospheres can be *approximated* as blackbodies, but a blackbody is a highly idealized construct; the actual spectra depart from it in bands and lines, which is where the information about composition lies.

The two sides of the Planck function

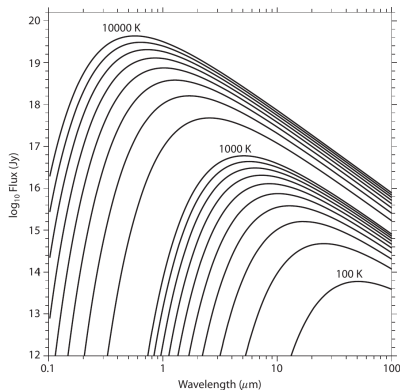


Figure 5: As in the previous slide; note the common slope on the long-wavelength side. Seager (2010), Fig. 2.7.

- **Rayleigh–Jeans side** ($h\nu \ll k_B T$, $\lambda \gg \lambda_{\text{max}}$): $B \simeq 2\nu^2 k_B T / c^2$. On this logarithmic plot it is a straight line of slope -2 in λ (slope $+2$ in ν), with the temperature entering only *linearly*. This is why the mid-infrared planet-to-star contrast is $\propto T_p / T_*$ rather than exponentially small (Lectures 1 and 4).
- **Wien side** ($h\nu \gg k_B T$): $B \simeq (2h\nu^3 / c^2) e^{-h\nu / k_B T}$, exponentially sensitive to T . A planet's flux at wavelengths shortward of its peak is negligible, which is why planets are invisible in the optical except by reflected light.
- The **peak** moves as $\lambda_{\text{max}} \propto 1/T$ (Wien's law, next slide); the **total** flux grows as T^4 (Stefan–Boltzmann, Section 3.3).

Wien's law: where does a blackbody peak?

Setting $\partial B(T, \lambda)/\partial \lambda = 0$ or $\partial B(T, \nu)/\partial \nu = 0$ (exercise 2.6) gives **Wien's displacement law** in the two forms

$$\lambda_{\max} T = 2898 \mu\text{m K}$$

$$\frac{\nu_{\max}}{T} = 5.88 \times 10^{10} \text{ Hz K}^{-1}$$

The two peaks do *not* represent the same spectral location: $\lambda_{\max} \nu_{\max} \neq c$, because B_{λ} and B_{ν} are densities per unit wavelength and per unit frequency, and $d\nu/d\lambda = -c/\lambda^2$ redistributes the weight. (Wien derived the law from thermodynamics; the numerical constants come from the transcendental equations $x = 5(1 - e^{-x})$ and $x = 3(1 - e^{-x})$, with $x = hc/\lambda k_B T$ or $h\nu/k_B T$.)

object	T (K)	$\lambda_{\max}$ of B_{λ}	$c/\nu_{\max}$ of B_{ν}
Sun (G2)	5750	$0.50 \mu\text{m}$	$0.88 \mu\text{m}$
M star	3500	$0.83 \mu\text{m}$	$1.46 \mu\text{m}$
hot Jupiter	1500	$1.9 \mu\text{m}$	$3.4 \mu\text{m}$
Earth	300	$9.7 \mu\text{m}$	$17 \mu\text{m}$
Jupiter	125	$23 \mu\text{m}$	$41 \mu\text{m}$

Consequences of Wien's law

- ▶ The Sun's B_λ peaks in the green; its B_ν peaks in the near-infrared. Neither is “the” color of the Sun; the choice of variable is a convention. When a paper quotes “the peak of the spectrum,” check which variable is meant.
- ▶ Planets radiate in the thermal infrared (5–50 μm), stars in the visible and near-infrared: the two components of a planet's spectrum, reflected starlight and thermal emission, are well separated in wavelength, except for the hottest planets ($T \gtrsim 2000\text{ K}$), whose thermal emission reaches into the near-infrared and even the red optical.
- ▶ Detecting a planet's thermal emission means observing near its Wien peak or beyond: 3–10 μm for hot Jupiters (Spitzer, JWST), 10–25 μm for temperate terrestrial planets (the 10 μm window for Earth analogs).
- ▶ Conversely, the stellar spectrum at the planet's thermal wavelengths is on the Rayleigh–Jeans tail, faint and smooth, which is what makes secondary eclipses measurable.
- ▶ Wien's law also sets the *photochemistry*: an M dwarf with $\lambda_{\text{max}} \simeq 0.8\text{--}1\text{ }\mu\text{m}$ emits much less near-UV than the Sun, so photolysis of H_2O , CH_4 , and O_3 proceeds differently on its planets, although flares and the chromospheric far-UV can compensate.

The Lambert surface: a surface of equal brightness

A **Lambert surface** scatters intensity *isotropically* (equally in all directions). It is the standard idealization of the reflectivity of planetary bodies, and it clarifies the difference between the intensity *emanating* from an object and the intensity *measured* by a distant observer.

- ▶ “Equal intensity in all directions” means the apparent surface brightness of an area element is the same from every viewing angle.
- ▶ Example: a sheet of white paper illuminated from above appears equally bright from any direction. Likewise, the spot cast by a light bulb on a white wall maintains its surface brightness when viewed from different angles.
- ▶ Contrast with a mirror (specular reflection) or with ice crystals that strongly backscatter: the brightness then depends strongly on the viewing direction.

We now show that a plane Lambert surface, illuminated along its normal, has the same apparent brightness from any direction, and see *why* this is consistent with fewer photons leaving at slant angles.

Two observers of a Lambert surface

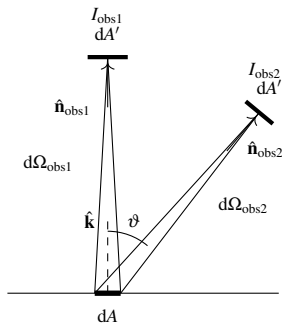


Figure 6: Intensity seen by two distant observers. Each collects the energy from dA through the area dA' set perpendicular to its own line of sight $\hat{n}_{\text{obs}}$, so the solid angle $d\Omega = dA'/\ell^2$ subtended by dA is smaller at observer 2 by $\cos \vartheta$. After Seager (2010), Fig. 2.8.

Consider a plane Lambert surface illuminated along its normal. Observer 1 looks along the normal, observer 2 at an angle ϑ from it; both view the same differential area dA from a distance much larger than the size of the surface element. From the definition of intensity,

$$I_{\text{obs1}} = \frac{dE_{\text{obs1}}}{dA' d\Omega_{\text{obs1}}},$$

$$I_{\text{obs2}} = \frac{dE_{\text{obs2}}}{dA' d\Omega_{\text{obs2}}}.$$

Here, dA' is the area perpendicular to $\hat{n}_{\text{obs1}}$ and $\hat{n}_{\text{obs2}}$. By the definition of the coordinate system, $\hat{n}_{\text{obs1}} \cdot \hat{k} = 1$ and $\hat{n}_{\text{obs2}} \cdot \hat{k} = \cos \vartheta$.

- **Geometry:** the solid angle subtended by dA at observer 2 is smaller than at observer 1 by the factor $\cos \vartheta$ (it approaches zero as $\vartheta \rightarrow 90^\circ$, when the element is seen edge-on).

$$d\Omega_2 = \frac{\cos \vartheta dA}{\ell^2} = \cos \vartheta d\Omega_1$$

Why a Lambert surface looks equally bright from all angles

- **Lambert's cosine law:** for isotropic scattering the energy leaving per unit *projected* area is the same in every direction, so the energy (or number of photons) emerging from dA toward observer 2 drops as $\cos \vartheta$ away from the normal:

$$dE_{\text{obs2}} = \cos \vartheta \, dE_{\text{obs1}}$$

Combining these two results, observer 2 measures

$$I_{\text{obs2}} = \frac{dE_{\text{obs1}} \cos \vartheta}{dA' d\Omega_{\text{obs2}}} \quad \Rightarrow \quad I_{\text{obs1}} = I_{\text{obs2}}$$

by comparison with the expression for observer 1.

- For a plane Lambert surface the intensity, i.e. the *brightness*, appears the same from all directions: the smaller number of photons emerging in the slant direction is exactly compensated by the smaller subtended angle of the area element.
- The *flux* received by observer 2 does decrease as $\cos \vartheta$; only the intensity is invariant. This is why a Lambert disk viewed obliquely looks just as bright but appears foreshortened.
- The same reasoning applied to a *sphere* gives the geometric albedo $2/3$ and the phase curve of a Lambert sphere (Chapter 3): each surface element is Lambertian, but the elements near the limb are lit obliquely, so they scatter fewer photons per unit area.

[Radiative Transfer] Start with the optical depth

“It is possible to sit through an entire radiative transfer class and not develop an intuition for the subject at all” (Heng). Rather than derive the transfer equation first, we introduce the one quantity needed to understand it: the **optical depth** τ .

- ▶ It quantifies whether a medium is transparent or opaque, and therefore which approximation describes the passage of radiation.
- ▶ It explains why the size of a star or a planet varies with wavelength, identifies the surface from which photons reach our telescopes, and defines the transit radius.

Clouds in the sky: some large clouds are transparent, some small clouds are opaque. Physical size alone is a poor indicator; two more ingredients matter: how tightly the material is packed (the number density n) and how absorbent each particle is (the cross section σ):

$$\tau \equiv \int n \sigma dx$$

a macroscopic quantity times a microscopic one, integrated along the path. A photon travels, on average, until it has traversed $\tau \sim 1$: τ is roughly the number of interactions the photon suffers. Transparent clouds have $\tau \ll 1$ (no interaction on average); opaque clouds $\tau \gg 1$ (many absorptions and reemissions). Walking through fog or snow, visibility extends to $\tau \sim 1$.

What $\tau \sim 1$ explains

- ▶ **Why the Sun is yellow, not an X-ray source.** The core is at 10^6 – 10^7 K (X-rays by Wien's law), but it sits at $\tau \gg 1$. What we see is the *photosphere*, the surface where $\tau \sim 1$ as measured from outside: the surface of last absorption or scattering before the photons stream to us.
- ▶ **Why the Sun has different sizes.** Its “radius” is the $\tau \sim 1$ surface, and since τ is wavelength dependent, the Sun observed in X-rays, UV, visible, or radio has different radii. (Styrofoam is opaque in the visible but transparent to microwaves.)
- ▶ **How deep starlight penetrates a hot Jupiter.** To the $\tau \sim 1$ surface at that wavelength: the concept applies symmetrically to emission from an object and absorption by it.
- ▶ **The transit radius.** During transit the measured radius picks out the chord across the limb where the *chord* optical depth is ~ 1 (next slide).
- ▶ **Which regime.** For $\tau \ll 1$ radiation is freely streaming and propagates at c ; for $\tau \gg 1$ it is absorbed and reemitted many times over a short distance and its passage resembles *diffusion*. The two-stream approximation (Heng Chapter 3) is designed for $\tau \lesssim 1$; the diffusion approximation for $\tau \gg 1$.

Photospheric radius versus transit radius

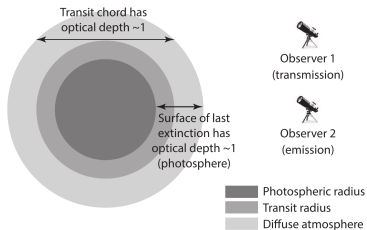


Figure 7: Observer 1 measures the transit radius (transit chord with $\tau \sim 1$); observer 2 sees photons from the photosphere, the surface of last extinction with $\tau \sim 1$ toward the observer. The transit radius exceeds the photospheric radius, exaggerated here. Reproduced from Heng (2017), Fig. 2.1.

Two observers, two radii, both defined by $\tau \sim 1$:

- **Emission (observer 2):** the photosphere is where the *vertical* optical depth from the top is ~ 1 .
- **Transmission (observer 1):** the transit radius is where the *chord* (slant) optical depth is ~ 1 .

For an isothermal hydrostatic atmosphere with scale height H (Heng, problem 2.8.6), along a chord at impact height $z = 0$ with $x^2 \simeq 2Rz$,

$$\tau_{\text{chord}} = n_{\text{ref}} \sigma \sqrt{2\pi HR}, \quad \tau_{\text{vert}} = n_{\text{ref}} \sigma H,$$

so $\tau_{\text{chord}}/\tau_{\text{vert}} = \sqrt{2\pi R/H}$: for a hot Jupiter ($R/H \sim 150$) about 30. The chord reaches $\tau \sim 1$ at lower density, i.e. *higher* altitude, by $\Delta z = H \ln \sqrt{2\pi R/H} \simeq 3.4H$. Hence the transit radius is larger than the photospheric radius, and transmission spectra probe lower pressures than emission spectra.

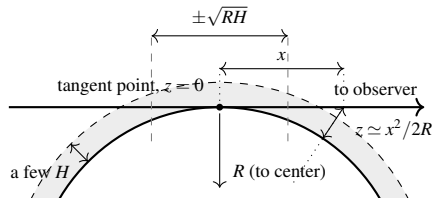
Where $\tau_{\text{chord}} = n_{\text{ref}} \sigma \sqrt{2\pi HR}$ comes from

Take an isothermal, hydrostatic atmosphere, $n(z) = n_{\text{ref}} e^{-z/H}$, and a ray that grazes the planet with its lowest point at the reference level $z = 0$. Measuring x along the ray from that point, the altitude of the ray is

$$z(x) = \sqrt{R^2 + x^2} - R \simeq \frac{x^2}{2R} \quad (x \ll R),$$

so the number density falls off as a *Gaussian* along the path, $n = n_{\text{ref}} e^{-x^2/2RH}$. Integrating,

$$\tau_{\text{chord}} = \sigma \int_{-\infty}^{\infty} n dx = n_{\text{ref}} \sigma \int_{-\infty}^{\infty} e^{-x^2/2RH} dx = n_{\text{ref}} \sigma \sqrt{2\pi HR}, \quad \tau_{\text{vert}} = \sigma \int_0^{\infty} n dz = n_{\text{ref}} \sigma H.$$



The two integrals differ only in their effective path length: H vertically, $\sqrt{2\pi HR}$ along the chord — the geometric mean of H and $2\pi R$.

The ray climbs as $x^2/2R$, so it leaves the dense layers once $|x| \gtrsim \sqrt{RH}$: that narrow stretch around the tangent point contributes almost all of the absorption, and everything further along the chord adds little.

Why the transit radius sits $\Delta z = H \ln \sqrt{2\pi R/H}$ higher

Let z be the tangent height of a chord, measured from the level where the *vertical* optical depth is unity, $n_{\text{ref}}\sigma H = 1$. Since $n(z) = n_{\text{ref}}e^{-z/H}$, the previous slide gives

$$\tau_{\text{chord}}(z) = n(z) \sigma \sqrt{2\pi R H} = \sqrt{\frac{2\pi R}{H}} e^{-z/H}.$$

A chord grazing the photospheric level is therefore already optically thick by the factor $\sqrt{2\pi R/H}$. Setting $\tau_{\text{chord}}(\Delta z) = 1$,

$$\Delta z = H \ln \sqrt{\frac{2\pi R}{H}} = \frac{H}{2} \ln \frac{2\pi R}{H}$$

- ▶ The starlight is blocked out to this height, so the **transit radius** is $R + \Delta z$, above the photosphere that sets the *emission* radius.
- ▶ Hot Jupiter ($R/H \simeq 150$): $\Delta z \simeq 3.4H$. With $H \simeq 400$ km that is $\simeq 1400$ km, which raises the transit depth by $2\Delta z/R \simeq 4\%$. Earth ($R/H \simeq 760$): $\Delta z \simeq 4.2H \simeq 35$ km.
- ▶ The dependence is only *logarithmic*, which is why Δz is a few H for objects of very different size — and why a transmission spectrum has features of order H in altitude.

The random walk of a photon (Heng, problem 2.8.2)

Photons created at the center of the Sun must travel $R_{\odot} \simeq 7 \times 10^5$ km to escape, moving a mean free path l_{mfp} between interactions.

- ▶ **Transparent Sun:** escape time $R_{\odot}/c \simeq 2.3$ s.
- ▶ **Mildly opaque** ($1 \lesssim \tau \lesssim 10$): the number of interactions is $N \sim R_{\odot}/l_{\text{mfp}} \sim \tau$.
- ▶ **Very opaque:** the photon random-walks. Its mean displacement is zero, but the mean *square* displacement after N steps is Nl_{mfp}^2 . Setting $\sqrt{N}l_{\text{mfp}} = R_{\odot}$ gives $N \sim (R_{\odot}/l_{\text{mfp}})^2 = \tau^2$.
- ▶ With $n \sim 10^{24} \text{ cm}^{-3}$ and Thomson scattering $\sigma \sim 10^{-24} \text{ cm}^2$: $l_{\text{mfp}} \sim 1 \text{ cm}$, $\tau \sim 7 \times 10^{10}$, $N \sim 5 \times 10^{21}$, and the escape time $Nl_{\text{mfp}}/c \sim 10^{11} \text{ s} \sim$ a few thousand years. (The real answer, 10^4 – 10^5 yr, differs because the density is far from uniform and absorption plus reemission is not a pure random walk — the key weakness of the estimate.)

The same reasoning tells us that in an optically thick atmosphere radiative energy transport is *diffusive* (e.g., Rosseland mean opacity).

Three ways to write the optical depth

There are three measures of how absorbent a medium is, hence three equivalent definitions of τ . *Extinction* means absorption plus scattering.

quantity	symbol	units (cgs)	optical depth
cross section (per particle)	σ	cm^2	$\tau = \int n \sigma dx$
opacity (cross section per unit mass)	κ	$\text{cm}^2 \text{g}^{-1}$	$\tau = \int \kappa d\tilde{m} = \int \kappa \rho dx$
extinction coefficient (per length)	$\alpha_e (\equiv \kappa \text{ in Seager})$	cm^{-1}	$\tau = \int \alpha_e dx$

- ▶ $\tilde{m}$ is the **column mass** (mass per unit area). In an atmosphere the pressure is the weight of the overlying column, $P = \tilde{m}g$ (Newton's second law per unit area), so $\tau = \int \kappa dP/g$: for constant κ , $\tau = \kappa P/g$, which is why pressure is the natural vertical coordinate of atmospheric radiative transfer.
- ▶ The cross section may be visualized as the target area presented by an atom, molecule, or particle.
- ▶ The reciprocal of the extinction coefficient is the **mean free path**, $l_{\text{mfp}} = 1/\alpha_e = 1/n\sigma$.
- ▶ **Terminology warning.** Some references call σ , κ , and α_e collectively the “extinction coefficient” although their units differ. Seager uses κ for the coefficient in m^{-1} and κ_m for the mass-independent opacity in $\text{m}^2 \text{kg}^{-1}$; Heng uses α_e and κ respectively. Always check the units.

Opacity: the most important atmospheric property

The most important physical quantity in the atmosphere that affects the transfer of radiation is the **opacity**: how opaque a substance is, how hard it is for radiation to pass through it.

- ▶ Opacity depends on the number density of particles and on their absorbing, scattering, and emitting properties, which in turn depend on temperature, pressure, and frequency.
- ▶ To describe radiative transfer we must therefore define coefficients for the absorption and emission of radiation. **The word “opacity” refers to the absorption and scattering (i.e. extinction) coefficients** of the atmospheric particles.
- ▶ For now the underlying microscopic processes of photon absorption and emission are hidden in these macroscopic coefficients; Chapter 8 (and Heng Chapter 5) supply the microscopic details: line lists, line profiles, Rayleigh scattering, condensates.
- ▶ Many different symbols are used for the opacity coefficients in other textbooks and in the literature (see the table of three definitions above); keep track of the units.

The **monochromatic extinction coefficient** $\kappa(\mathbf{x}, \nu)$ [m^{-1}] is defined by the energy removed from a beam of radiation in a volume dV and solid angle $d\Omega$, per unit time and frequency:

$$dE = \kappa(\mathbf{x}, \nu) I(\mathbf{x}, \hat{\mathbf{n}}, \nu) dV d\Omega d\nu dt$$

Absorption and scattering

κ includes all processes that remove energy from the beam:

$$\kappa(\mathbf{x}, \nu) = \alpha(\mathbf{x}, \nu) + \sigma_s(\mathbf{x}, \nu)$$

- ▶ **True absorption** α : the photon is destroyed. For example, a photon is absorbed by a molecule (dust, cloud particles), and the excitation energy is then converted to kinetic energy of the gas (dust, clouds) by a subsequent collision. The photon's energy has joined the thermal pool.
- ▶ **Scattering** σ_s : the photon is removed from the beam by a change of direction (and possibly a change of energy). The subscript s distinguishes the scattering coefficient σ_s [m^{-1}] from the cross section σ [m^2]. Examples: Rayleigh scattering by molecules, Mie scattering by cloud particles.
- ▶ Extinction is *isotropic* in a static medium: the absorbing or scattering particle does not care from which direction the photon comes, so κ has no $\hat{\mathbf{n}}$ dependence. (In a moving medium the Doppler shift introduces an angle dependence, which we ignore for atmospheres.)
- ▶ The distinction between α and σ_s is physical, not bookkeeping: absorbed energy heats the gas and is reemitted thermally at the local temperature; scattered energy keeps its frequency and carries no information about the local temperature.

The emission coefficient: thermal emission and scattering

The **monochromatic emission coefficient** $\varepsilon(\mathbf{x}, \hat{\mathbf{n}}, \nu)$ [$\text{J m}^{-3} \text{sr}^{-1} \text{s}^{-1} \text{Hz}^{-1}$] describes the energy emitted into a volume dV within a solid angle $d\Omega$ per unit time and frequency,

$$dE = \varepsilon(\mathbf{x}, \hat{\mathbf{n}}, \nu) dV d\Omega dt d\nu.$$

Planetary-atmosphere emission includes both thermal emission and scattering, because both processes add to the beam.

- **Thermal emission**, by the Kirchhoff–Planck relation (*Kirchhoff's law*, valid in local thermodynamic equilibrium, Section 5.4):

$$\varepsilon_{\text{therm}}(\mathbf{x}, \nu) = \alpha(\mathbf{x}, \nu) B(\mathbf{x}, \nu)$$

where B is the Planck function at the local temperature. Thermal emission is isotropic in a static medium: atoms and molecules have no preferred direction of radiation.

- **Scattering** of radiation arriving from all directions $\hat{\mathbf{n}}'$ into the beam direction $\hat{\mathbf{n}}$:

$$\varepsilon_{\text{scat}}(\mathbf{x}, \hat{\mathbf{n}}, \nu) = \sigma_s(\mathbf{x}, \nu) \frac{1}{4\pi} \oint_{\Omega'} P(\Theta) I(\mathbf{x}, \hat{\mathbf{n}}', \nu) d\Omega'$$

with the scattering angle $\cos \Theta = \hat{\mathbf{n}} \cdot \hat{\mathbf{n}}'$; primes denote the direction of incidence. We consider only *coherent* scattering, in which the photon retains its frequency.

The single-scattering phase function

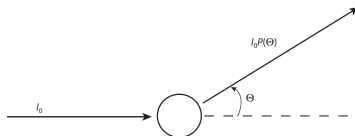


Figure 8: The scattering angle Θ used in $P(\Theta)$: incident intensity I_0 , scattered intensity $I_0 P(\Theta)$. Seager (2010), Fig. 5.1.

Scattering can be anisotropic, so we define the dimensionless **single-scattering phase function** $P(\Theta)$: the redirection of the incident intensity into the outgoing intensity, i.e. the 3D directional scattering probability. It is normalized so that

$$\frac{1}{4\pi} \oint_{\Omega} P(\Theta) d\Omega = 1$$

- If the scattering is isotropic, $P(\Theta) = 1$ and the scattering emissivity reduces to $\epsilon_{\text{scat}}(\mathbf{x}, \nu) = \sigma_s(\mathbf{x}, \nu) J(\mathbf{x}, \nu)$, with J the mean intensity of Chapter 2: an isotropic scatterer simply reemits the mean intensity it sees.
- Θ is the angle between the incident direction $\hat{\mathbf{n}}'$ and the scattered direction $\hat{\mathbf{n}}$, $\cos \Theta = \hat{\mathbf{n}} \cdot \hat{\mathbf{n}}'$; for a spherically symmetric particle P depends only on Θ , not on the absolute orientation.

Phase functions in practice

- ▶ **Rayleigh scattering** by molecules has $P(\Theta) = \frac{3}{4}(1 + \cos^2 \Theta)$: symmetric between forward and backward, mildly anisotropic (Chapter 8).
- ▶ **Cloud and haze particles** comparable to or larger than the wavelength have strongly forward-peaked P (Mie theory, Chapter 8), often approximated by the Henyey–Greenstein function with asymmetry parameter $g_0 = \langle \cos \Theta \rangle$; the two-stream treatment of Heng Chapter 3 uses only g_0 .
- ▶ The origin of the phase function is microscopic: the angular pattern of dipole radiation for Rayleigh scattering, and the interference of waves scattered from different parts of a large particle for Mie scattering.
- ▶ P must not be confused with the *illumination* phase function Φ_α used for phase curves (Section 3.6): both are called “phase function” for historical reasons, but P describes a single scattering event and Φ_α the brightness of the whole planet versus orbital phase. The relation between the two, through multiple scattering in the atmosphere, is what a reflected-light model computes.

Two points to remember about scattering: it is both a *source* and a *sink* for the beam — as a source it counts as emission, as a sink as extinction; and it is angle dependent, unlike isotropic thermal emission, which is why the emission coefficient carries an $\hat{\mathbf{n}}$.

Number densities, cross sections, and level populations

The absorption and scattering coefficients are sums of number density times cross section over all species j and all their internal states i :

$$\alpha(T, P, \nu) = \sum_j \alpha_j(T, P, \nu) = \sum_j \sum_i n_{ji}(T, P) \sigma_{ji}(T, P, \nu)$$

and similarly $\sigma_s = \sum n \sigma_{\text{scat}}$ for Rayleigh scattering by molecules and for solid particles.

- ▶ T and P vary with location, so (T, P) and $\mathbf{x}$ are interchangeable arguments.
- ▶ The number densities come from the elemental abundances, atmospheric chemistry, and escape (Chapter 4); the partition among states i from statistical mechanics (LTE, below).
- ▶ The inputs are enormous: for water vapor, *hundreds of millions* of lines from transitions among electronic, vibrational, and rotational states (Heng Chapter 5: the “million- to billion-line challenge”).

Mass-independent opacity. Many books use $\kappa_m(T, P, \nu) = \sum_j \kappa_j / \rho_j$ [$\text{m}^2 \text{kg}^{-1}$], the extinction per unit mass of species j (Heng’s κ). Its virtue: the number density depends strongly on T and P through the ideal gas law, so $\kappa = n\sigma$ varies enormously with height, whereas $\kappa_m = \kappa/m$ varies only through the cross section itself (line broadening, level populations), much more slowly.

Extinction by particles: the extinction efficiency

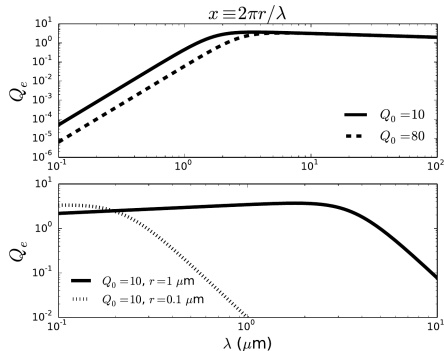


Figure 9: Extinction efficiency of spherical cloud particles versus normalized radius $x = 2\pi r/\lambda$ (top) and versus wavelength for $r = 0.1$ and $1 \mu\text{m}$ (bottom). $Q_0 = 10$ and 80 mimic refractory and volatile compositions. Reproduced from Heng (2017), Fig. 2.2.

A spherical particle of radius r has geometric cross section πr^2 , but its actual cross section for extinction is generally larger,

$$\sigma = Q_e \pi r^2, \quad Q_e = Q_a + Q_s$$

with the **extinction efficiency** Q_e , separable into absorption and scattering parts.

- Its functional form and value depend on whether the particle is “small” or “large”. “The beauty about physics is that whether something is small or large is not a subjective judgment based on human emotion”: for the interaction of radiation with matter, it is a comparison with the wavelength, through the normalized radius

$$x \equiv \frac{2\pi r}{\lambda}.$$

- A particle is small if $x \ll 1$ and large if $x \gg 1$.

Computing Q_e

- ▶ To derive the exact functional form of $Q_e(\lambda, r)$ requires a full-blown calculation of absorption and scattering by the particle: Mie theory for homogeneous spheres, given the complex refractive index $n(\lambda) + ik(\lambda)$ of the material, or numerical methods (T-matrix, discrete dipole approximation) for nonspherical particles.
- ▶ In practice, atmospheric scientists look up tables of $Q_e(\lambda, r)$ (and of ω_0 and g_0) and apply them to the models. Optical constants exist for silicates, iron, corundum, water, ammonia, methane, and other candidate condensates (<https://github.com/seoncafe/SEDust>).
- ▶ Roughly speaking, the extinction efficiency has a somewhat universal form,

$$Q_e \sim \min \left\{ \left(\frac{2\pi r}{\lambda} \right)^4, 1 \right\},$$

with the transition from Rayleigh scattering ($x \ll 1$) to scattering by large particles ($x \gg 1$) occurring at a value of the normalized radius that depends on composition.

- ▶ With the exception of resonant transitions specific to each material (e.g. the $10\text{ }\mu\text{m}$ silicate feature), the overall shape of Q_e is similar for different members of the same family, e.g. the olivine group.

A fitting formula for Q_e

- ▶ A crude but useful **fitting formula** (Heng Eq. 2.25):

$$Q_e \simeq \frac{5}{Q_0 x^{-4} + x^{0.2}} \quad x = \frac{2\pi r}{\lambda},$$

which asymptotes to a roughly constant value for $x \gg 1$ and to Rayleigh scattering ($Q_e \propto x^4$) for $x \ll 1$.

- ▶ The dimensionless parameter Q_0 is a proxy for composition: refractory species (e.g. silicates) tend to have $Q_0 \approx 10$, while volatile ones (e.g. water) have $Q_0 \approx 40\text{--}80$. A larger Q_0 shifts the Rayleigh-to-geometric transition to larger x .
- ▶ The top panel of Heng Fig. 2.2 holds for *all* particle radii: as a function of the normalized radius x , the extinction efficiency is size independent to lowest order. The bottom panel shows the same curve mapped to wavelength for $r = 0.1$ and $1\text{ }\mu\text{m}$.
- ▶ Given Q_e , a cloud with number density n_c of particles of radius r contributes an extinction coefficient $\kappa_c = n_c Q_e \pi r^2$, to be added to the gas opacity in the radiative transfer; for a size distribution $n_c(r)$ one integrates over r .

Small and large particles

- ▶ $x \ll 1$: **Rayleigh regime.** $Q_e \propto x^4$, i.e. $\sigma \propto r^6/\lambda^4$: small particles scatter blue light strongly and are transparent in the infrared. This is the origin of the blue sky, of the Rayleigh slope in transmission spectra, and of the “haze” signature at short wavelengths.
- ▶ $x \gg 1$: **geometric regime.** $Q_e \rightarrow$ a roughly constant value ($\simeq 2$ for total extinction, the “extinction paradox”: geometric blocking plus an equal amount of diffraction); the particle is *grey*, affecting all wavelengths equally.
- ▶ A micron-sized particle is radiatively *large* for visible light but *small* in the far-infrared. Generally the influence of a cloud particle on absorption and scattering is felt only at $\lambda \lesssim 2\pi r$, which generalizes straightforwardly to a cloud with a size distribution.
- ▶ The particle radius exerts a somewhat larger influence on Q_e than the composition (bottom panel of Heng Fig. 2.2).
- ▶ **Consequence for spectra:** a cloud of $\sim 1 \mu\text{m}$ particles adds a grey continuum opacity in the optical and near-infrared, flattening the transmission spectrum and muting molecular features, while leaving the mid-infrared unaffected. Sub-micron hazes instead produce a steep, Rayleigh-like slope.

Single-scattering albedo and asymmetry factor

Distinguishing the absorption and scattering cross sections, the **single-scattering albedo** is

$$\omega_0 \equiv \frac{\sigma_s}{\sigma_s + \sigma_a} \quad (\text{Seager's } \tilde{\omega}, \text{ and in coefficient form } \omega_0 = \sigma_s/\kappa),$$

the fraction of light scattered rather than absorbed in a single interaction: $\omega_0 = 1$ is pure (conservative) scattering, $\omega_0 = 0$ pure absorption.

The second quantity needed to describe a scattering event is the **scattering asymmetry factor** $g_0 = \langle \cos \Theta \rangle$, the phase-function-weighted mean cosine of the scattering angle (Heng Chapter 3):

- ▶ $g_0 = 0$: symmetric scattering (isotropic, or Rayleigh, which is symmetric fore-aft but not isotropic);
 $g_0 > 0$: forward peaked; $g_0 < 0$: backscattering.
- ▶ Asymptotically $g_0 \rightarrow 0$ for $2\pi r/\lambda \ll 1$ and $g_0 \rightarrow 1$ for $2\pi r/\lambda \gg 1$: small particles scatter isotropically, large particles deflect radiation mostly forward. A sub-micron particle scatters infrared light isotropically but forward-scatters X-rays.

Both $\omega_0(\lambda, r)$ and $g_0(\lambda, r)$ are tabulated from Mie calculations for each composition. An atmosphere contains an enormous number of particles of different species; collectively they set the fraction of light reflected at each wavelength, i.e. the albedos of last lecture. Heng Chapter 4 relates A_B to ω_0 and g_0 in the two-stream approximation.

Mean opacities: why a plain average is useless

In many treatments of radiative transfer we want a *mean* opacity so that we can work with equations averaged over frequency (grey treatments, Heng Chapter 4). At first thought, one might take the literal average of the opacity over all frequencies. In practice this would not be useful.

- ▶ Consider a planet radiating approximately as a blackbody, with exceedingly high opacity at UV frequencies but close to zero opacity everywhere else. A plain average would be dominated by the UV, where there is virtually no planetary flux; a better representation of this atmosphere would simply be zero opacity.
- ▶ The point is that we are interested in how much *flux* is blocked by, or allowed through, the atmosphere at different frequencies. Useful mean opacities are therefore weighted by a function of the intensity.
- ▶ Because the intensity of the planet atmosphere is unknown — it is the quantity we are trying to solve for — the blackbody intensity is used as the weight.

Planck mean, weighted by B :

$$\kappa_P(T, P) = \frac{\int_0^\infty \kappa(T, P, \nu) B(T, \nu) d\nu}{\int_0^\infty B(T, \nu) d\nu}$$

Where the opacity is high, the contribution to the mean is high. **The Planck mean is valid in optically thin regimes**: it gives the total thermal emission of a thin layer correctly, since $\int \alpha_\nu B_\nu d\nu = \kappa_P \sigma_R T^4 / \pi$.

The Rosseland mean

Rosseland mean, a *harmonic* mean weighted by the temperature derivative of the Planck function:

$$\frac{1}{\kappa_R(T, P)} = \frac{\int_0^\infty \kappa(T, P, \nu)^{-1} [\partial B(T, \nu) / \partial T] d\nu}{\int_0^\infty [\partial B(T, \nu) / \partial T] d\nu}$$

- ▶ It gives a *higher* weight to frequencies with *small* opacity than to frequencies with large opacity, capturing the physical situation that more radiation travels through the atmosphere at the frequencies where the opacity is smallest — the spectral windows. This is the opposite of the Planck mean.
- ▶ It follows from *radiative diffusion* (Chapter 6): deep in an optically thick atmosphere the flux at each frequency is $F_\nu = -\frac{4\pi}{3\kappa_\nu} \frac{\partial B_\nu}{\partial T} \frac{dT}{dz}$, and integrating over frequency gives $F = -\frac{16\sigma_R T^3}{3\kappa_R} \frac{dT}{dz}$: the total flux is carried where $1/\kappa_\nu$ is large, so the harmonic mean is the right average.
- ▶ The weight $\partial B/\partial T$ peaks at slightly shorter wavelengths than B itself ($\lambda T \simeq 2400 \mu\text{m K}$), because the Wien side responds more strongly to a temperature change.

Typically the Planck mean is valid in optically thin regions of the atmosphere and **the Rosseland mean in optically thick regimes**; neither is right in between, which is why grey models are only qualitative and why Heng Chapter 5 discusses other means and the k -distribution method.

Optical depth, formally, and the plane-parallel convention

From our viewpoint distance is measured in km; for photons the natural distance scale is the optical depth, a measure of transparency. Along a path s ,

$$d\tau(s, \nu) = -\kappa(s, \nu) ds$$

dimensionless; $\tau \ll 1$ optically thin (a photon typically crosses without interacting), $\tau > 1$ optically thick. Because κ depends on frequency and position, so does $\tau(\mathbf{x}, \nu)$.

- In **plane-parallel** atmospheres τ is measured *backward* along the outgoing ray: $\tau = 0$ at the top of the atmosphere, increasing downward, because the observer looks down into the atmosphere. Hence the minus sign:

$$\tau(z, \nu) = - \int_{z_{\max}}^z \kappa(z', \nu) dz' = \int_z^{z_{\max}} \kappa(z', \nu) dz'.$$

We write τ_ν with the z dependence implied.

- Later we will see that τ is the dimensionless *e-folding* scale for the attenuation of hot radiation through a cooler gas layer.
- Seager's κ is Heng's α_e ; in mass units, $d\tau = \kappa_m \rho ds$, and with hydrostatic equilibrium $d\tau = \kappa_m dP/g$.

The mean free path

Consider a slab of area A^2 , thickness dz , and number density n , each particle presenting a cross section σ . The probability that an incident photon is stopped in the slab is the ratio of the net area of the targets to the slab area,

$$P(z) = \frac{n\sigma A^2 dz}{A^2} = n\sigma dz,$$

which is the same as the fractional drop of the beam intensity that we have already described, $dI = -\kappa I dz = -n\sigma I dz$. This suggests a reasonable definition of the **mean free path**, the mean distance a photon travels before it is stopped by an interaction with a molecule or other atmospheric particle:

$$l = \frac{1}{n\sigma} = \frac{1}{\kappa}, \quad d\tau = \frac{dz}{l}$$

- ▶ The optical depth counts the path length in units of mean free paths; $\tau \sim 1$ means “about one interaction”, as Heng’s intuitive argument said.
- ▶ Typical values: in Earth’s lower atmosphere at visible wavelengths (Rayleigh scattering, $\sigma \simeq 5 \times 10^{-27} \text{ cm}^2$, $n \simeq 2.5 \times 10^{19} \text{ cm}^{-3}$) $l \simeq 80 \text{ km}$, so the clear sky has $\tau \simeq 0.1$ vertically; inside a water cloud l is tens of meters; in the $15 \mu\text{m}$ CO_2 band core l is meters.

The mean free path, more formally

Start from $dI = -\kappa I dz$, i.e.

$$\frac{dI}{I} = -\frac{dz}{l} \quad \Longrightarrow \quad I = I_0 e^{-z/l}.$$

The probability that a photon is absorbed between z and $z + dz$ is the fraction of the initial beam removed there,

$$dP(z) = \frac{I(z) - I(z + dz)}{I_0} = \frac{1}{l} e^{-z/l} dz,$$

a properly normalized probability distribution ($\int_0^\infty dP = 1$). Its expectation value is

$$\langle z \rangle = \int_0^\infty z dP(z) = \int_0^\infty \frac{z}{l} e^{-z/l} dz = l$$

The mean distance traveled before an interaction is indeed $l = 1/\kappa$, which justifies the name.

- In optical-depth units the distribution is $dP = e^{-\tau} d\tau$, with mean $\langle \tau \rangle = 1$ and standard deviation 1: interaction distances are broadly distributed, and a sizable fraction of photons travel $\tau > 2$ or 3 without interacting ($e^{-2} = 14\%$, $e^{-3} = 5\%$).
- Monte Carlo radiative transfer samples exactly this distribution: $\tau = -\ln \xi$ with ξ a uniform random number in $(0, 1]$, then the photon is moved to the point where the accumulated optical depth along its direction equals τ .

Local thermodynamic equilibrium (LTE)

Complete thermodynamic equilibrium requires thermal, chemical, and mechanical equilibrium — impossible across an atmosphere with huge gradients of T and P and an open boundary at the top. **Local** thermodynamic equilibrium (LTE) applies in a local region where the gradients are small compared to the photon mean free path.

- ▶ In LTE we assume all the conditions of thermodynamic equilibrium hold for the *matter*, but we let the *radiation field* depart from a blackbody. The radiation field in a planetary atmosphere is very different from B ; it must be computed from the transfer equation.
- ▶ Why it matters: the coefficients κ and ϵ are sums of number densities times cross sections over all **level populations** (the numbers of atoms or molecules in each quantum state). We need (1) the overall number densities and (2) the fraction in each state.
- ▶ The heart of the problem: the level populations are determined by the radiation field (photoexcitation, stimulated emission, photoionization) — the very quantity we are solving for. A full simultaneous solution of the radiation field and the populations is possible in principle but onerous in practice.
- ▶ **LTE decouples them:** all properties of the matter depend only on the local kinetic temperature and one other variable (ρ or P). Level populations follow the Boltzmann distribution, ionization the Saha equation, molecular abundances chemical equilibrium (Chapter 8 and Heng Chapter 7).

When is LTE valid?

LTE implies a strict connection between the matter and the local temperature; it is valid where that connection is physically realized.

- ▶ **High densities: collisions dominate radiative processes.** An electron in an atom or molecule is excited to a higher level by absorbing a photon. Later, because of a collision, it cascades downward, releasing the energy as kinetic energy of the colliding particles; after further elastic collisions this energy ends up in the thermal pool of the gas. In this case of collisional deexcitation the photon is said to be *destroyed* and converted into kinetic energy. In this way collisions couple the photons (the radiation field) to the matter temperature via the kinetic energy of the gas. (As long as the colliding particles have a Maxwellian velocity distribution, the level populations have their Boltzmann equilibrium values.)
- ▶ **A blackbody radiation field.** If the mean radiation field is a blackbody, $J(\mathbf{x}, \nu) = B(\mathbf{x}, \nu)$, the radiative rate equations that control the level populations also give an equilibrium (Boltzmann) distribution (Section 8.2.4). This second situation is not really relevant for planetary atmospheres, and from the first example it is clearly not a *necessary* condition for LTE.
- ▶ The criterion is a comparison of rates: LTE holds for a transition when the collisional deexcitation rate exceeds the radiative one, $n_{\text{coll}} q_{ul} \gg A_{ul}$, i.e. above a *critical density* that depends on the transition (Chapter 8).

LTE in Earth's atmosphere and in exoplanets

- ▶ **Earth:** LTE is actually valid from the ground up to about 60 km, largely because collisional rates between molecules dominate radiative rates, driving a Boltzmann distribution. We know from spectral measurements at the top of the atmosphere and at different altitudes that Earth's radiation field is *not* a blackbody — LTE and blackbody radiation are different things. Above ~ 60 km (mesosphere, thermosphere) the CO_2 $15\ \mu\text{m}$ band and other bands are in non-LTE: collisions are too rare to keep the vibrational levels thermalized.
- ▶ **Exoplanets:** LTE is commonly used in exoplanet atmosphere calculations, and it is a good approximation for the pressures probed by emission spectra ($\gtrsim 1$ mbar). But there are regions where it is not valid: the very upper layers where radiation escapes freely through the open boundary. There the radiation field is very different from a blackbody *and* collisions do not dominate over radiative transition rates, causing a departure from LTE. Examples: CH_4 fluorescence at $3.3\ \mu\text{m}$ in giant planets, the escaping upper atmospheres probed by UV transits, and the thermospheres heated by stellar XUV.
- ▶ **Validation:** for a given situation one could compare radiative and collisional rates, but to accurately validate LTE a non-LTE calculation itself may be necessary. In practice one checks whether the pressures probed by the observation are above the critical densities of the relevant transitions.

LTE and Kirchhoff's law

Kirchhoff's law of radiation states that in thermal equilibrium the emissivity of a body equals its absorptivity:

$$\epsilon_{\text{therm}}(\mathbf{x}, \nu) = \alpha(\mathbf{x}, \nu) B(\mathbf{x}, \nu)$$

- ▶ The thermal emission at $\mathbf{x}$ is blackbody radiation weighted by the absorptive properties of the gas: any time a photon is absorbed by gas in LTE, the energy is reemitted as radiation, with amount and wavelength set by the Planck function times the absorption coefficient. Gas that absorbs strongly at a frequency also emits strongly there — spectral lines appear in absorption or emission depending on the temperature gradient, not on the opacity alone.
- ▶ Why it must hold (exercise 5.1): inside a blackbody enclosure at temperature T the radiation field is B and is unchanged by the presence of any body at the same T ; the transfer equation $dI/ds = -\kappa I + \epsilon$ with $I = B$ constant then demands $\epsilon = \kappa B$ for every frequency and direction.
- ▶ Strictly valid only in complete thermodynamic equilibrium. In planetary atmospheres we justify it for small, localized regions where the gradients of the thermodynamic properties are small compared to a photon mean free path.
- ▶ Note what Kirchhoff's law does *not* say: it does not make $I = B$. The emissivity is tied to the local temperature; the intensity is whatever the transfer equation produces.

The source function

The **source function** (sometimes called the contribution function) is the ratio of the emission coefficient to the extinction coefficient:

$$S(\mathbf{x}, \hat{\mathbf{n}}, \nu) = \frac{\varepsilon(\mathbf{x}, \hat{\mathbf{n}}, \nu)}{\kappa(\mathbf{x}, \nu)}$$

It has the same dimensions as the intensity, it simplifies the transfer equation and its solution, and it provides physical insight: it is the intensity that the beam *relaxes toward* as it travels through the medium (a beam brighter than S is attenuated, a beam fainter than S is filled in).

Expand the source function into a thermal-emission component and a scattering component. With Kirchhoff's law $\varepsilon_{\text{therm}} = \alpha B$ and isotropic coherent scattering $\varepsilon_{\text{scat}} = \sigma_s J$, under conditions of LTE,

$$S(\mathbf{x}, \nu) = \frac{\alpha(\mathbf{x}, \nu) B(\mathbf{x}, \nu) + \sigma_s(\mathbf{x}, \nu) J(\mathbf{x}, \nu)}{\alpha(\mathbf{x}, \nu) + \sigma_s(\mathbf{x}, \nu)} \iff \boxed{S = (1 - \omega_0) B + \omega_0 J}$$

with the single-scattering albedo $\omega_0 = \sigma_s / (\alpha + \sigma_s)$: a weighted mean of the Planck function and the mean intensity.

The source function: limits and consequences

- ▶ **Pure absorption** ($\omega_0 = 0$): the strict LTE value $S = B$, from Kirchhoff's law with no scattering. The beam relaxes toward the local Planck function; the emergent spectrum then reflects the temperature structure directly.
- ▶ **Pure scattering** ($\omega_0 = 1$): $S = J$, the mean intensity. The beam relaxes toward the local radiation field itself; the medium neither creates nor destroys photons, it only redistributes their directions.
- ▶ The scattering term makes S depend on the radiation field: the transfer equation becomes an *integro-differential* equation for I . Even the expression with $\omega_0 \neq 0$ “allows a slight deviation from LTE” in Seager's words: the level populations are still thermal, but the source function is not B .
- ▶ In general, while $I = B$ for blackbody radiation, LTE radiation is *not* necessarily blackbody radiation: in LTE, I is not required to equal B . LTE constrains the *matter* (Kirchhoff, Boltzmann, Saha); the *radiation* is whatever the transfer equation produces.
- ▶ **Convention check.** Heng writes $S = \omega_0 J / 4\pi + (1 - \omega_0)B$ because his $J \equiv \oint I d\Omega$ is 4π times Seager's mean intensity $J = \frac{1}{4\pi} \oint I d\Omega$. Same physics; check which J a paper uses before comparing formulas.

The radiative transfer equation along a ray

Radiative transfer describes how a beam changes as it travels a distance s through a volume of gas: losses κI and gains ε . For a static atmosphere,

$$\frac{dI(\mathbf{x}, \hat{\mathbf{n}}, \nu)}{ds} = -\kappa(\mathbf{x}, \nu) I(\mathbf{x}, \hat{\mathbf{n}}, \nu) + \varepsilon(\mathbf{x}, \hat{\mathbf{n}}, \nu)$$

Time-dependent derivation. In an inertial frame, follow the energy in a fixed volume element of length ds and cross section dA during dt ; for moving material κ and ε acquire an angle dependence through the Doppler shift. Balancing losses and gains,

$$[I(\mathbf{x} + \Delta\mathbf{x}, \hat{\mathbf{n}}, \nu, t + \Delta t) - I(\mathbf{x}, \hat{\mathbf{n}}, \nu, t)] dA d\Omega d\nu dt = [-\kappa I + \varepsilon] ds dA d\Omega d\nu dt,$$

and with $\Delta t = \Delta s/c$ the left side is $(\frac{1}{c}\partial_t I + \partial_s I)ds$, giving

$$\left(\frac{1}{c}\frac{\partial}{\partial t} + \frac{\partial}{\partial s}\right) I(\mathbf{x}, \hat{\mathbf{n}}, \nu, t) = -\kappa(\mathbf{x}, \hat{\mathbf{n}}, \nu, t) I + \varepsilon(\mathbf{x}, \hat{\mathbf{n}}, \nu, t).$$

- ▶ The ∂_t/c term matters only when conditions change on light-crossing times; for atmospheres we drop it (static case), together with the $\hat{\mathbf{n}}$ dependence of κ , while remembering that radiation is an energy *flow* per unit time.
- ▶ In vacuum ($\kappa = \varepsilon = 0$) the equation says I is constant along the ray, as claimed last lecture.

Heng's shortest derivation, in optical depth

Since τ is the correct measure of transparency, work directly with it. Let the intensity impinge on the surface of an object at angle θ from the normal, $\mu \equiv \cos \theta$, with the convention that reflection (upward travel) has $\mu > 0$ and penetration into the object $\mu < 0$.

- ▶ Loss of intensity by extinction across the object's optical depth $\Delta\tau$ (measured along the normal): $I\Delta\tau/\mu$ (the slant path is $\Delta\tau/|\mu|$ long).
- ▶ Compensated by the object's own emission: $-S\Delta\tau/\mu$.

$$\Delta I = \frac{\Delta\tau}{\mu} (I - S) \quad \xrightarrow{\Delta\tau \rightarrow 0} \quad \boxed{\mu \frac{\partial I}{\partial \tau} = I - S}$$

- ▶ “It looks deceptively simple, because all of the complexity has been hidden within the source function.”
- ▶ The same equation follows from Seager's form by substituting $d\tau = -\kappa dz$, $dz = \mu ds$, and $S = \epsilon/\kappa$ (next slides). The sign convention (τ increasing downward, $\mu > 0$ upward) is the standard one for atmospheres.
- ▶ Historical note: the equation was first derived for stars by astrophysicists (Schuster, Schwarzschild, Milne, Chandrasekhar) and later adopted by the atmospheric and climate sciences.

The plane-parallel approximation

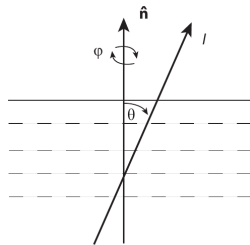


Figure 10: A 1D plane-parallel atmosphere: stratified layers, vertical coordinate z , ray at angle θ from the normal $\hat{n}$, azimuth ϕ . Reproduced from Seager (2010), Fig. 5.2.

The plane-parallel atmosphere is a good framework in which to study the radiative transfer equation.

- ▶ For a 1D planar atmosphere, the atmosphere is defined by a stratified plane with each layer having homogeneous properties such as T , P , and ρ . This 1D plane represents one location on the surface of the planet.
- ▶ **When is it valid?** If the radial depth of the atmosphere is much smaller than the planetary radius. The photosphere spans a few scale heights: $H/R_p \sim 0.01$ for hot Jupiters, 10^{-3} for Earth, so the curvature of the layers is negligible for rays that are not nearly horizontal.
- ▶ The plane-parallel definition assumes **axial symmetry** about the vertical: there is no preferred horizontal direction in the medium itself (the illumination may still come from a specific direction).
- ▶ The vertical coordinate z increases upward; the optical depth τ increases downward from $\tau = 0$ at the top.

The plane-parallel transfer equation

Considering the axial symmetry and adopting z as the (1D) vertical coordinate,

$$\frac{\partial}{\partial x} = \frac{\partial}{\partial y} = 0 \quad \text{and} \quad \frac{dz}{ds} = \cos \theta \equiv \mu,$$

where θ is the angle between the surface normal and the beam of intensity. Substituting into the equation along a ray,

$$\mu \frac{dI(z, \hat{\mathbf{n}}, \nu)}{dz} = -\kappa(z, \nu) I(z, \hat{\mathbf{n}}, \nu) + \varepsilon(z, \hat{\mathbf{n}}, \nu)$$

If we further consider axial symmetry of the radiation field, i.e. no ϕ dependence,

$$\mu \frac{dI(z, \mu, \nu)}{dz} = -\kappa(z, \nu) I(z, \mu, \nu) + \varepsilon(z, \mu, \nu),$$

and a single angle variable $\mu \in [-1, 1]$ describes the direction: $\mu > 0$ upward, $\mu < 0$ downward, $\mu = 0$ horizontal.

- ▶ The factor μ is the whole geometric content of the approximation: a slant ray traverses $ds = dz/\mu$ of path for each dz of height, so it sees $1/\mu$ times more optical depth.
- ▶ Horizontal rays ($\mu \rightarrow 0$) travel infinitely far in a plane-parallel atmosphere; in reality they leave the curved atmosphere after a path $\sim \sqrt{2\pi R_p H}$, the transit chord.

The scattering integral and the limits of plane-parallel geometry

As written, the plane-parallel equation is an *integro-differential* equation, because $\varepsilon = \varepsilon_{\text{scat}} + \varepsilon_{\text{therm}}$ and the scattering term contains angle integrals of I :

$$\varepsilon_{\text{scat}}(z, \mu, \nu) = \sigma_s(z, \nu) \frac{1}{4\pi} \int_0^{2\pi} \int_{-1}^1 P(\mu, \phi; \mu', \phi') I(z, \mu', \nu) d\mu' d\phi'.$$

Even with azimuthal symmetry of the medium, the phase function couples every direction to every other direction, so the equation for $I(\mu)$ at one angle involves I at all angles.

Where the plane-parallel picture fails.

- ▶ **Transmission at the terminator:** the starlight travels along a chord, and the geometry is inherently spherical; the slant path length $\sqrt{2\pi R_p H}$ comes from the curvature.
- ▶ **Grazing incidence near the limb** of the dayside, where the stellar zenith angle approaches 90° and $1/\mu_0$ diverges; spherical (or pseudo-spherical) corrections are needed.
- ▶ **Extended atmospheres** inflated by escape or by very low gravity, where H/R_p is no longer small.
- ▶ **Horizontal inhomogeneity:** day–night contrasts, patchy clouds, and the eclipse-mapping problem require 3D or at least a collection of 1D columns.

Moments and the equation in optical depth

In plane-parallel geometry with azimuthal symmetry the mean intensity and flux of Chapter 2 become

$$J(z, \nu) = \frac{1}{4\pi} \int_0^{2\pi} \int_{-1}^1 I d\mu d\phi = \frac{1}{2} \int_{-1}^1 I(z, \mu, \nu) d\mu, \quad F(z, \nu) = \int_0^{2\pi} \int_{-1}^1 \mu I d\mu d\phi = 2\pi \int_{-1}^1 \mu I(z, \mu, \nu) d\mu.$$

(The stellar-atmosphere literature uses $H = F/4\pi$ because its form parallels J and K .)

Now change to the optical-depth scale ($d\tau_\nu = -\kappa dz$) and use the source function:

$$\mu \frac{dI(\tau_\nu, \mu, \nu)}{d\tau_\nu} = I(\tau_\nu, \mu, \nu) - S(\tau_\nu, \mu, \nu)$$

the same form Heng obtained directly. Note the roles of the two hemispheres: **outgoing** rays have $\mu > 0$, **incoming** rays $\mu < 0$.

Boundary conditions.

- Top ($\tau_\nu = 0$): the stellar radiation incident on the atmosphere, $I(0, \mu, \nu) = I_*(0, \mu_0, \nu)$ for $-1 \leq \mu \leq 0$, a plane-parallel beam at the single angle μ_0 . For the nightside, for infrared calculations where the planet's own emission dominates, or for unirradiated planets, $I_* = 0$.
- Bottom ($\tau_{\max, \nu}$): the intensity from the planet's interior, $I(\tau_{\max, \nu}, \mu, \nu) = I_{\text{int}}$ for $0 \leq \mu \leq 1$, matched to the interior energy flux (or, for a terrestrial planet, the surface emission and reflection).

The formal solution

Multiply the equation by the integrating factor $e^{-\tau_v/\mu}$:

$$\frac{dI}{d\tau_v} e^{-\tau_v/\mu} - \frac{I}{\mu} e^{-\tau_v/\mu} = \frac{d}{d\tau_v} \left(I e^{-\tau_v/\mu} \right) = -\frac{S}{\mu} e^{-\tau_v/\mu}.$$

Integrating from an initial optical depth $\tau_{v,i}$ to a final one $\tau_{v,f}$,

$$I(\tau_{v,f}, \mu, \nu) = I(\tau_{v,i}, \mu, \nu) e^{-(\tau_{v,i} - \tau_{v,f})/\mu} - \frac{1}{\mu} \int_{\tau_{v,i}}^{\tau_{v,f}} S(\tau_v, \mu, \nu) e^{-(\tau_v - \tau_{v,f})/\mu} d\tau_v$$

- **First term:** the initial intensity diminished by the exponential attenuation of absorption over the optical path $(\tau_i - \tau_f)/\mu$.
- **Second term:** the emission from the atmosphere, an exponentially weighted average of the source function along the beam up to the location of interest.
- For an outgoing ray ($\mu > 0$) from deep inside ($\tau_i > \tau_f$) the integral runs downward in τ and the second term is positive: each layer contributes $S d\tau/\mu$, attenuated by $e^{-(\tau - \tau_f)/\mu}$ on its way up. For an incoming ray ($\mu < 0$) the same formula, read from the top down, describes the penetration of starlight with the atmosphere's emission added.

It is called the “formal” solution because S must be known; in general it is not.

The emergent intensity

Our goal is the planet's spectrum: the emergent flux at the top of the atmosphere, the measurable quantity. For a *semi-infinite* atmosphere, integrate from deep inside, $\tau_{v,i} = \infty$, to the top, $\tau_{v,f} = 0$, using $\lim_{\tau_v \rightarrow \infty} I e^{-\tau_v/\mu} = 0$ (the deep layers are hidden):

$$I(0, \mu, \nu) = \frac{1}{\mu} \int_0^\infty S(\tau_v, \mu, \nu) e^{-\tau_v/\mu} d\tau_v \quad (0 < \mu \leq 1).$$

The emergent intensity is the amount of intensity from each altitude that reaches the surface along a path at angle θ to the line of sight.

- The weight $e^{-\tau/\mu}/\mu$ is normalized ($\int_0^\infty e^{-\tau/\mu} d\tau/\mu = 1$) and peaks at the top, with mean $\langle \tau_v \rangle = \mu$: the emergent intensity is a weighted sample of S over $\tau_v \lesssim \mu$, the “photosphere” at that frequency and angle. Rays emerging at grazing angles ($\mu \rightarrow 0$) sample shallower, usually cooler layers — this is **limb darkening**.

Why the problem is hard

- **The source function depends on I through scattering.** $S = (1 - \omega_0)B + \omega_0 J$, or with the angle-dependent scattering emissivity

$$\varepsilon_{\text{scat}}(\tau, \mu, \nu) = \sigma_s(\tau, \nu) \frac{1}{4\pi} \int_0^{2\pi} \int_{-1}^1 P(\mu, \phi; \mu', \phi') I(\tau, \mu', \nu) d\mu' d\phi',$$

and J is an integral of the unknown I . Physically, scattering *decouples* the intensity from the local conditions: photons may scatter through large distances in the atmosphere without interacting with the thermal pool of the gas via absorption and thermal reemission. A numerical solution of an integro-differential equation is required.

- **Even B depends on I** through the temperature structure: radiative equilibrium ties $T(\tau)$ to the absorbed radiation (next section), so S is unknown until the whole problem, including the energy balance of every layer, is solved.

Take-home messages

Optical depth. $\tau = \int n\sigma dx = \int \kappa\rho dx = \int \alpha_e dx$; photons travel to $\tau \sim 1$; this defines photospheres, transit radii ($\tau_{\text{chord}} = n\sigma\sqrt{2\pi HR}$), and regimes (streaming versus diffusion, $N \sim \tau^2$). Mean free path $1/\kappa$.

Opacity and emission. $\kappa = \alpha + \sigma_s$ (isotropic); $\varepsilon = \alpha B + \sigma_s \frac{1}{4\pi} \oint PI d\Omega'$; particles: $\sigma = Q_e \pi r^2$, ω_0 , g_0 ; Planck and Rosseland means for thin and thick regimes. LTE: matter properties depend only on local T and P (collisions dominate); Kirchhoff $\varepsilon = \alpha B$; source function $S = (1 - \omega_0)B + \omega_0 J$.

Transfer equation. $\mu dI/d\tau = I - S$ in plane-parallel geometry; formal solution $I(0, \mu) = \frac{1}{\mu} \int_0^\infty S e^{-\tau/\mu} d\tau$; hard because S depends on I (scattering, radiative equilibrium) and the boundary conditions are split between top and bottom. Beer's law for pure extinction; isothermal layers exchange flux with $\mathcal{T} = 2E_3(\Delta\tau)$.

Next. Solving that equation: boundary conditions, Beer's law and the transmission function, radiative equilibrium, clouds and retrieval, and then line formation and the two-stream approximation (Lecture 5).

References

- ▶ Seager, S. 2010, *Exoplanet Atmospheres: Physical Processes* (Princeton University Press), Chapter 5.
- ▶ Heng, K. 2017, *Exoplanetary Atmospheres: Theoretical Concepts and Foundations* (Princeton University Press), Chapter 2.
- ▶ Mihalas, D. 1978, *Stellar Atmospheres*, 2nd ed. (Freeman) — the most thorough treatment of LTE.
- ▶ Rybicki, G. B. and Lightman, A. P. 1986, *Radiative Processes in Astrophysics* (Wiley), Chapter 1.
- ▶ Chandrasekhar, S. 1960, *Radiative Transfer* (Dover).
- ▶ Goody, R. M. and Yung, Y. L. 1989, *Atmospheric Radiation: Theoretical Basis*, 2nd ed. (Oxford).
- ▶ Pierrehumbert, R. T. 2010, *Principles of Planetary Climate* (Cambridge).
- ▶ Toon, O. B., et al. 1989, *JGR* **94**, 16287 (two-stream with linear source function).
- ▶ Rodgers, C. D. 2000, *Inverse Methods for Atmospheric Sounding* (World Scientific) — optimal estimation.
- ▶ Madhusudhan, N. and Seager, S. 2009, *ApJ* **707**, 24; Line, M. R., et al. 2013, *ApJ* **775**, 137; Benneke, B. and Seager, S. 2012, *ApJ* **753**, 100 — exoplanet retrieval.
- ▶ Lecavelier des Etangs, A., et al. 2008, *A&A* **481**, L83; Fortney, J. J. 2005, *MNRAS* **364**, 649 — chord optical depth and transit radius.